

Healthcare Data Analytics Project

A comprehensive analysis of hospital performance across multiple cities

By: Prabhav Sanjeev

Tools Used: Python | SQL | Power BI

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1. Introduction

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This project focuses on analyzing healthcare data to understand patient trends, revenue generation, and treatment costs across multiple cities.
- The main objective of this project is to extract meaningful insights using data analysis tools like Python, SQL, and Power BI.
- **Added Business Value:** The pipeline established here helps healthcare administrators optimize their operational workflows and financial strategies.

2. Dataset Description

The dataset used in this project is a healthcare dataset containing approximately 500 unique patient records, strictly verified using the `nunique()` function in Pandas to ensure absolute data consistency. It includes the following crucial columns for holistic hospital management:

Column Name	Description
Patient ID	Unique alphanumeric identifier for each patient record
Age	Chronological age of the admitted patient
City	Geographic location of the hospital branch
Disease	Primary clinical diagnosis upon admission
Admission Date	Timestamp of patient intake and registration
Treatment Cost	Base cost of medical procedures and hospitalization
Payment Status	Current clearance state of the bill (e.g., Paid, Unpaid)
Insurance Cover	Amount or percentage of the bill covered by insurance policies

3. Data Cleaning and Processing

- Data cleaning was meticulously performed using Python libraries such as Pandas and

NumPy.

- Missing values across clinical and financial records were handled programmatically to avoid skewing the analytical models.
- New analytical features were engineered to enrich the dataset, such as:
 - **Age Group:** Binned into demographic classes to track age-specific disease vulnerability.
 - **Final Bill Amount:** Calculated by factoring the Insurance Cover against the gross Treatment Cost.
- This crucial transformation step ensured that the dataset was ready for high-level enterprise analysis.

4. Data Analysis

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Data analysis was performed comprehensively using Python and SQL tracks. Python was leveraged for initial Exploratory Data Analysis (EDA) and structural mapping.
- Descriptive statistics were generated to understand the distribution of patient ages, average treatment costs, and disease prevalence.
- SQL was subsequently utilized to filter and extract meaningful, aggregated insights from the processed database table schemas.

5. SQL Integration

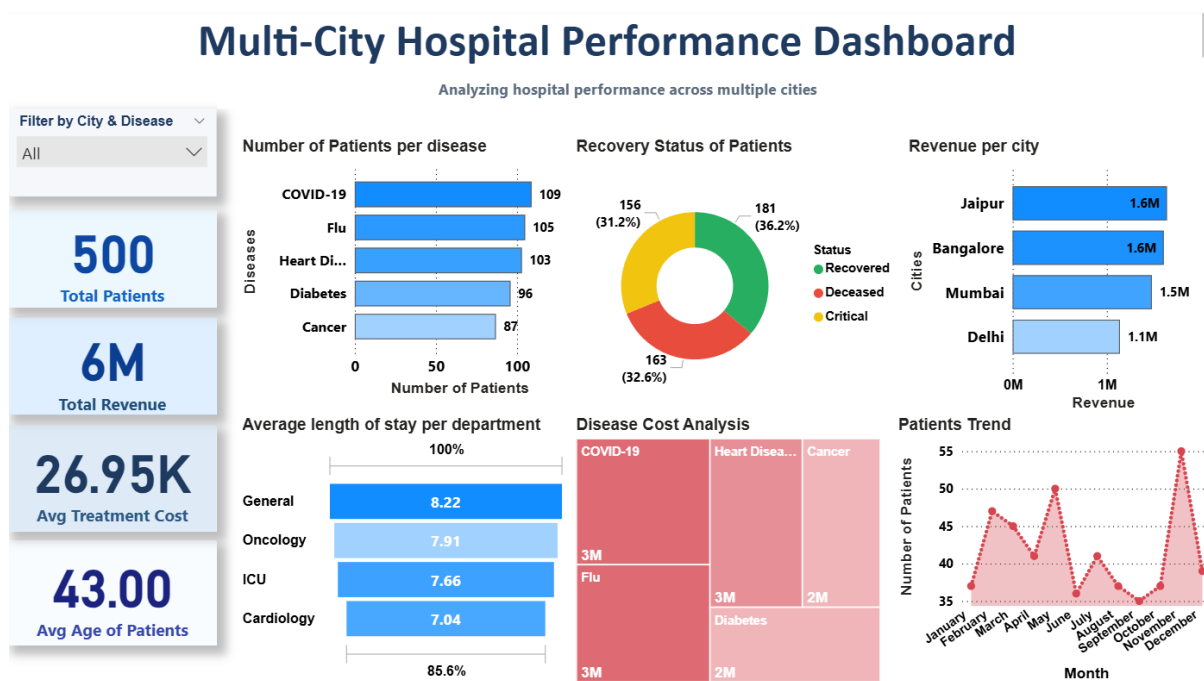
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The processed healthcare dataset was seamlessly connected to a MySQL relational database infrastructure using Python automation frameworks like SQLAlchemy and PyMySQL connectors.
- Complex SQL queries (utilizing JOINS, GROUP BY clauses, and window functions) were successfully executed to analyze the data further, including filtering performance metrics by city, tracking global disease distributions, and auditing payment statuses.

6. Data Visualization

An interactive dashboard was created using Power BI to visualize key performance indicators

(KPIs) such as revenue distribution by city, disease breakdowns, and operational patient trends, empowering executive stakeholders to consume complex business insights rapidly.



7. Key Insights

- **Geographical Disparity:** Certain cities contribute significantly higher revenue streams compared to others, indicating a potential operational need for smart resource reallocation.
- **Cost Drivers:** Some complex diseases (e.g., Oncology, Cardiology) are consistently associated with higher gross treatment costs.
- **Seasonal Variance:** Patient inflow varies robustly across different months, showing distinct seasonal peaks that require elastic administrative staffing models.
- **Revenue Leakage:** A visible portion of bills remains unpaid, indicating potential revenue optimization gaps and a direct need for stricter accounts receivable protocols.

8. Conclusion

- This project successfully demonstrated how structured data analytics methodologies can be effectively deployed in the healthcare sector to derive meaningful strategic insights.

- The end-to-end processing pipeline enhanced my operational understanding of applying appropriate data methodologies and tools (ETL transformations, Relational Databases, BI dashboards) to solve complex, real-world operational business problems.

9. Future Improvements

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Scale the analytics engine using larger, real-world global healthcare datasets for more accurate analysis and reduced statistical variance.
- Implement predictive Machine Learning modeling tracks to forecast macro patient admission trends and individual patient readmission risks.
- Deploy the executive reporting dashboards via a real-time data streaming architecture for live hospital performance monitoring.